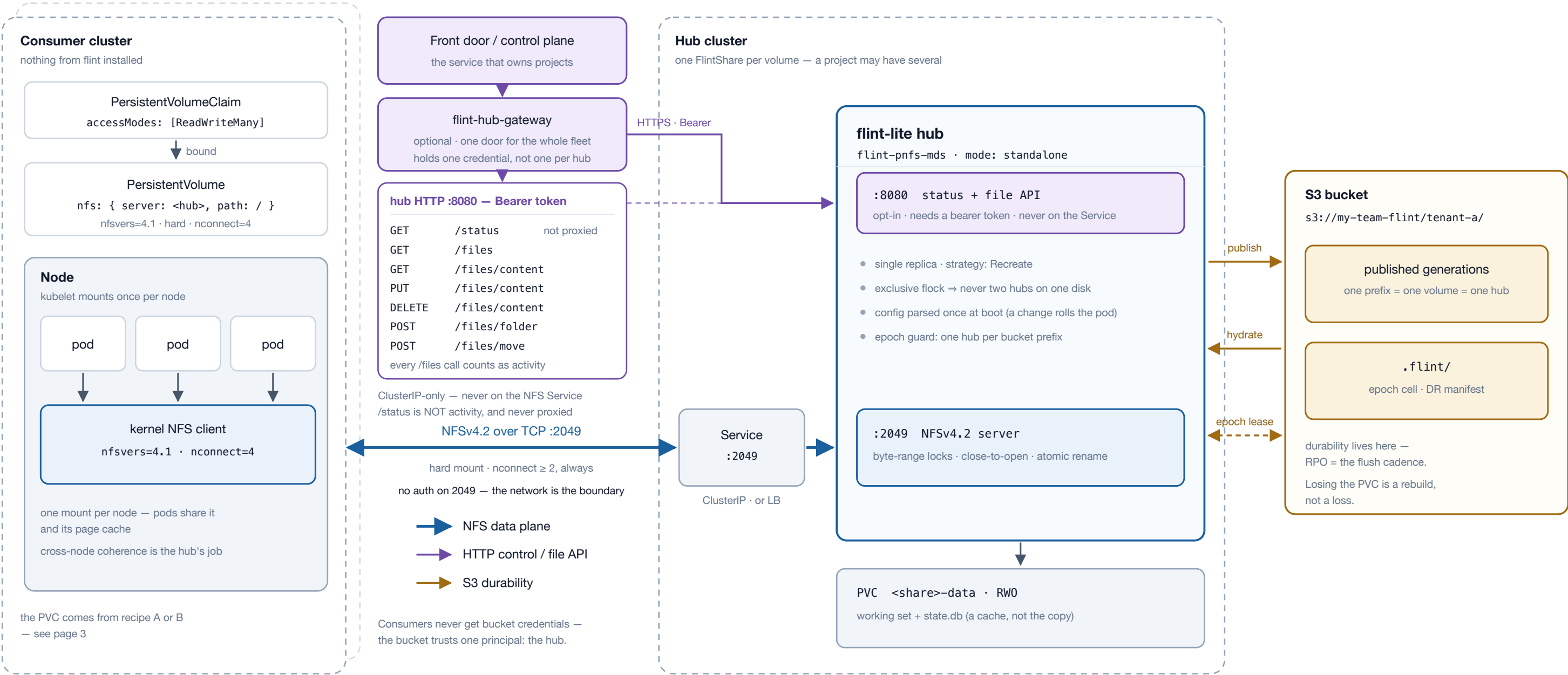
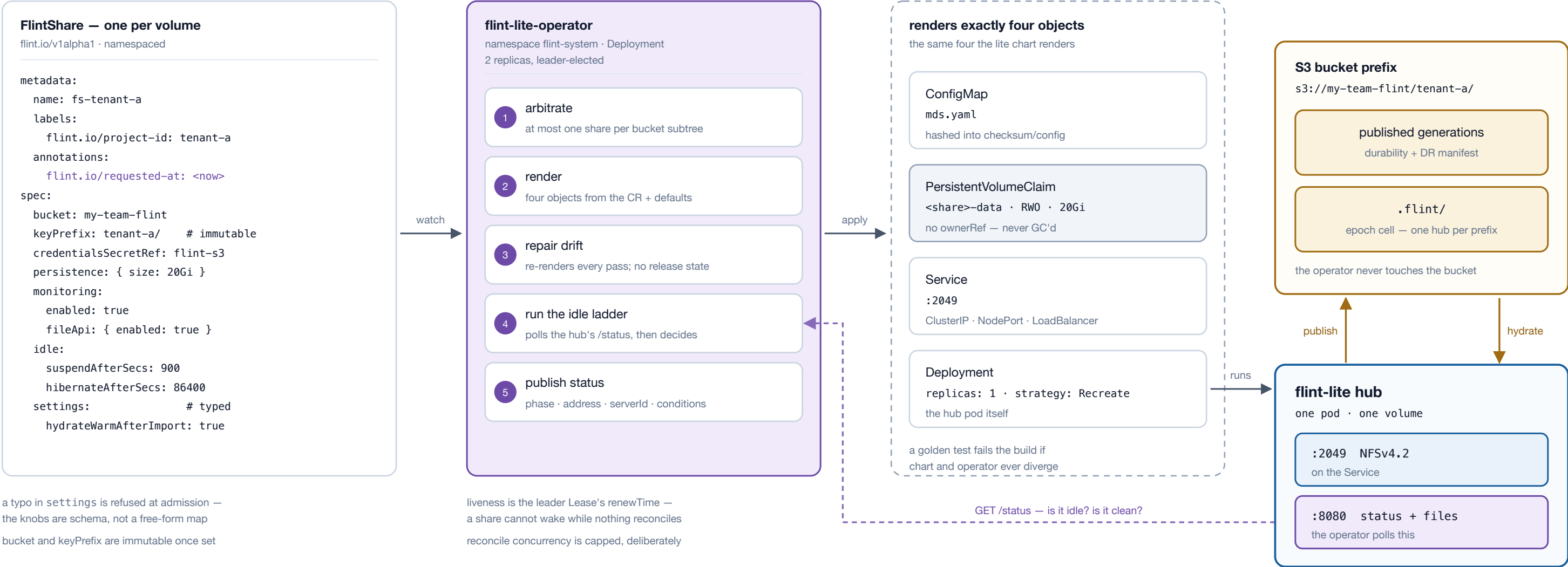


A single pod serves real POSIX over NFSv4.2. The PVC is the working set, an S3 prefix is the durability, and a second listener exposes the same filesystem over HTTP for callers that cannot mount — reached through one gateway rather than one address per hub. One hub serves exactly one prefix; a project that needs several volumes has several hubs.



One hub is the coherence authority for one tree — two hubs over one prefix is split-brain, which the store-side epoch lease fences and the operator refuses up front. **nconnect ≥ 2 is not optional:** without it the kernel opens exactly one connection and silently refuses every additional trunk — no error, on either side, ever. **The two doors are deliberately separate:** :2049 is on the Service and may become a LoadBalancer; :8080 can rewrite every file in the share, so it stays ClusterIP-only and never joins it. Each HTTP endpoint is an NFS compound dispatched in-process, not a second reader of the directory — so it inherits eviction handling, the write gate, symlink rules and the tier's capture notes. A cold file read through either door hydrates from the bucket first: NFS parks the caller with NFS4ERR_DELAY, HTTP answers 503 + Retry-After. **They are secured differently, and only one of them has a credential.** The file API needs a bearer token per share and mounts no routes at all without one — 404, not 401. Port 2049 is AUTH_SYS: the client asserts its own uid and the server takes it, so POSIX modes are enforced against a self-declared identity and **reachability is the boundary**. That control is networkPolicy in both charts — off by default, admitting nothing when both client lists are empty, and silently ignored by a CNI that does not enforce it.

One custom resource replaces one helm release, per hub. The operator re-renders the same four objects from the CR on every pass — there is no reusable release state to go stale, and nothing it does ever reaches the bucket.



Three things it will not do

- 1 · It never touches a bucket. Deleting a share deletes Kubernetes objects; S3 is exactly as durable as before.
- 2 · It never garbage-collects a PVC — the claim carries no ownerReference, deliberately.
- 3 · It will not run two hubs on one bucket subtree.

The fleet is one command

```
$ kubectl get flintshares -A
```

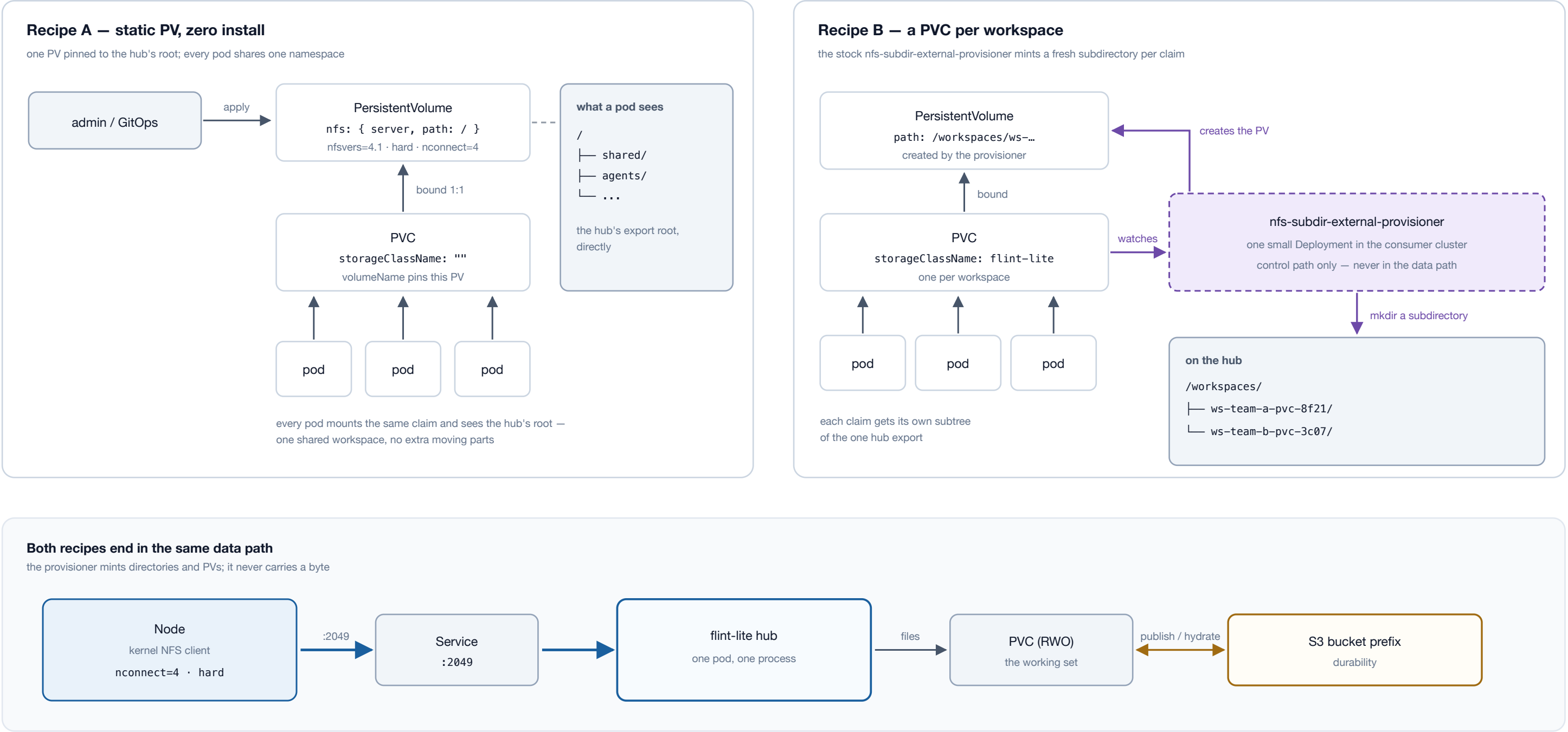
NAMESPACE	NAME	PHASE	ADDRESS
workspaces	tenant-a	Ready	tenant-a.workspaces.svc:2049
workspaces	tenant-b	Starting	
workspaces	tenant-c	IdleSuspended	

The front-door contract

```
GET flintshares/fs-<id> → 404? create
PATCH flint.io/requested-at: <now>
poll until .status.phase == Ready
read .status.address, .status.serverId
a changed serverId means a fresh PVC — remount
with a gateway in front, the middle three are one POST .../wake
```

Label the share flint.io/project-id, and flint.io/volume-id too once a project owns more than one — those labels are what the gateway resolves /v1/projects/<id>/volumes/<v> against, and volume-id defaults to the CR name. Deriving the name from the project id (fs-<project-id>) still works for a single-volume project and is what makes ensure-live idempotent: two front-door replicas racing a first touch issue the same create, one gets 409 AlreadyExists, and both proceed. **Uniqueness is enforced from the controller's cache**, not by the API server: Kubernetes cannot express "at most one share per (endpoint, bucket, prefix subtree)" in CEL, since both CEL and a ValidatingAdmissionPolicy see one object. Oldest wins; the loser goes Failed with a Conflict condition and gets no hub. **The project id is not part of that key**, and nothing in the operator reads it — which is precisely why one project may own several volumes: they collide only if their prefixes do. **That enforcement stops at the cluster boundary**. Two clusters each admit a share on one prefix and neither sees the other — and the store-side epoch only catches it when the prefixes are equal, because tenant-a/ and tenant-a/sub/ mint different epoch objects and never contend. Own prefix allocation upstream of Kubernetes.

Consumers install nothing from flint either way: the data path is the node kernel's NFS client. The choice is only who mints the PV — you, once, by hand; or a stock provisioner, once per workspace.



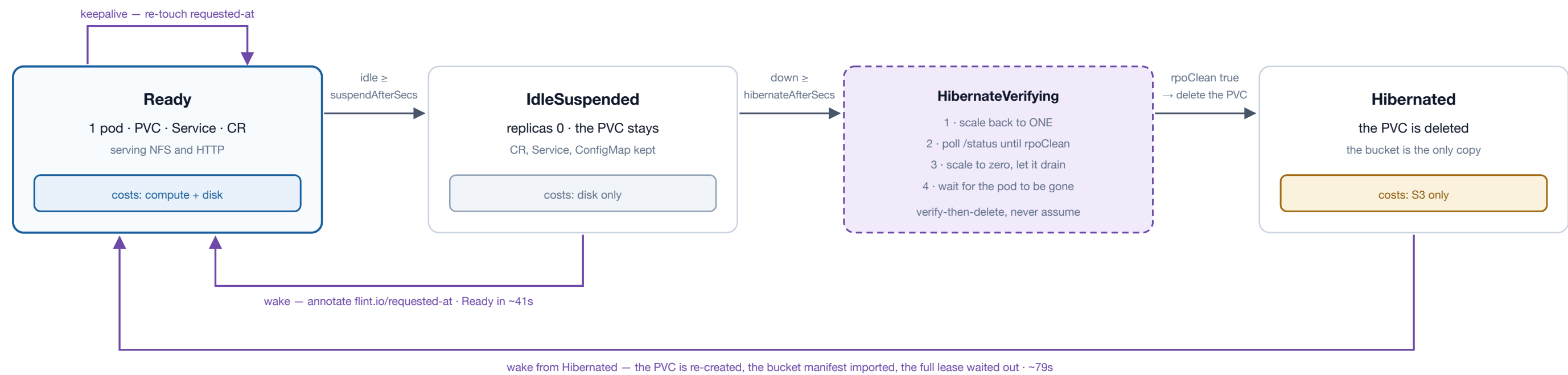
The same nconnect caveat applies to both recipes — recipe B passes mount options straight to the node's mount, so leaving it out gets one connection per node and no error saying so. **Recipe A is one shared tree**; recipe B is StorageClass semantics on top of it, so pick B when each workspace should be isolated and minted on demand, A when the point is that everybody sees the same files. Kubelet mounts once per node: pods co-scheduled on a node share one kernel client and one page cache, while cross-node and cross-cluster coherence — locks, close-to-open — is the hub's job. **Either way, a suspended hub is not woken by a mount.** A hard mount against a scaled-to-zero share hangs, because there are no endpoints and nothing to notice the attempt — which is why the wake annotation is written before an address is handed to anyone.

A tiered hub does real work before its listener binds, and the operator winds an unused share down in two rungs — each opt-in, each giving back something different. Nothing in the data path ever starts a hub.

Starting a hub — what happens before port 2049 opens



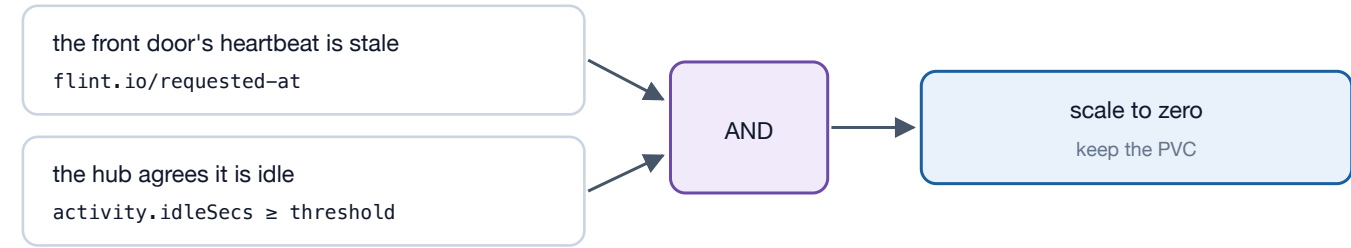
The idle ladder — each rung gives back something different



Nothing in the data path wakes a share: an NFS mount against a scaled-to-zero hub hangs, the file API answers 503 fast. Something must write the annotation — and a share cannot wake while no operator is reconciling.

spec.lifecycle: Suspended is an admin decision and a wake request does NOT override it — the phase reads Suspended, not IdleSuspended, so a front door can tell "will wake on request" from "never coming back".

Suspending needs two independent signals to agree



A hub that cannot be polled is never suspended — an unreachable hub is an unknown hub, not an idle one, and HubReachable says so.

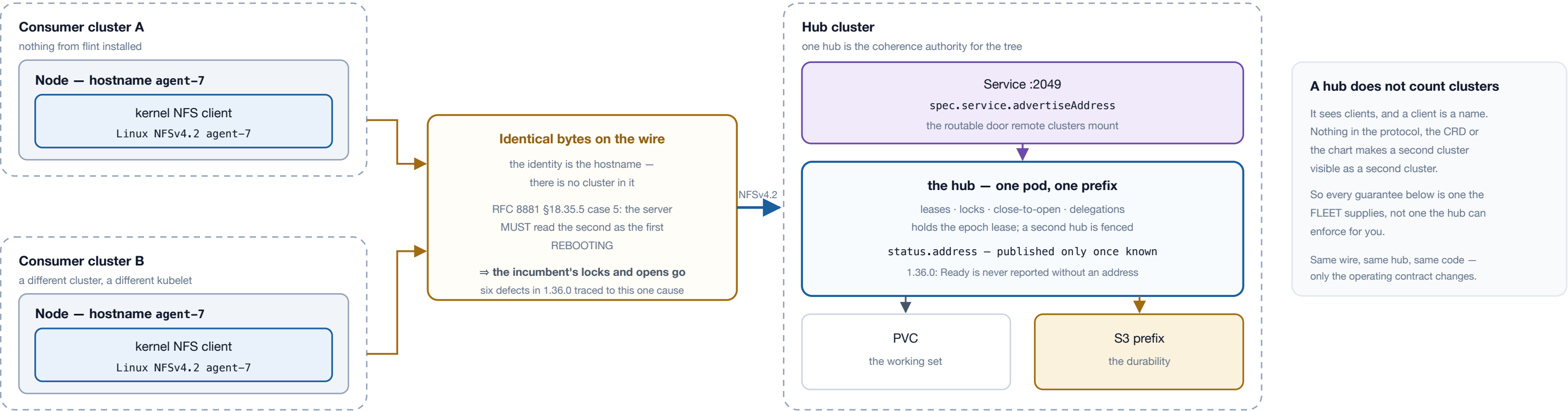
What each rung still holds

	CR	Svc + CM	PVC	Pod
Ready	held	held	held	held
IdleSuspended	held	held	held	released
Hibernated	held	held	released	released

Deleting the CR drops every object — an archived project costs zero Kubernetes objects. The bucket is never touched, under any policy.

The two rungs are not one setting with two numbers. Suspend keeps the disk, so waking is a pod start and an epoch re-claim on the same `state.db` — safe for any share, tiered or not. Hibernate deletes the PVC, so it requires `spec.bucket` and waking is a full DR import. **Hibernation is verified at drain time, not assumed:** the hub exits 0 whether or not it flushed and scale-to-zero destroys the pod, so the operator scales back to one and waits for `rpoClean` before it deletes anything. `rpoClean: null` — a share with no bucket — is a refusal, not a pass. **Hibernated wakes are slower than they look** (~79s against ~13s for the same-identity case): destroying the volume destroys the `serverId`, so self-recognition cannot short-circuit the lease and the claim waits out the full `lease_misses × heartbeat`. Every long-parked project pays that on its first open. **Do not keepalive from a liveness poller.** `/status` is deliberately not activity and the file API deliberately is — a UI that refreshes a listing on a timer pins its project awake forever, and a conditional GET answering `304` pins it exactly as hard.

One hub can serve agents in several clusters at once — the wire is ordinary NFSv4.2 and the hub does not know or care how many clusters are on it. What changes is that three things the protocol and the chart cannot supply become the fleet's responsibility: a unique name per client, a reachability boundary that is not `nfsClientCIDRs`, and a keepalive so a partitioned cluster's live agents are not mistaken for idle ones.



WHAT THE FLEET MUST SUPPLY — three things the protocol and the chart cannot do for you

1 · Give every client a unique name

The identity is the hostname and nothing else, so two clusters that share a node name are ONE client to the hub. Uniqueness has to hold across every cluster that mounts it — a fleet-wide property, not a per-cluster one.

Do either:

hostname = <cluster>--<node>

nfs.nfs4_unique_id=<cluster>

The second is a kernel module parameter set on the CLIENT node. It survives a rename and is the safer choice when you do not control node naming.

2 · Bound reachability at the network layer

Port 2049 is AUTH_SYS — the client asserts its own uid and the server takes it, so reachability IS the boundary.

But nfsClientCIDRs cannot draw it.

Nothing sets externalTrafficPolicy, so kube-proxy SNATs a NodePort or LB client to an address in the HUB's own cluster before the packet ever reaches the pod.

Measured on a three-cluster rig: 1486 of 1486

connections from two remote clusters arrived as the hub cluster's gateway — none as either remote node.

Listing remote CIDRs is an outage; listing the local one admits everyone. Use peering, SGs, or a gateway.

3 · Give a remote mount a keepalive

The idle ladder suspends a share on quiet, and a partitioned cluster's busy agents look exactly like quiet ones from the hub's side.

suspendWithSessions defaults to suspending anyway:

the protective value is false, and it is opt-in. Read the name literally.

Even set, it only NARROWS the window — leases EXPIRE, so a long enough partition drops the count to zero and the guard stops guarding.

Drive POST /wake every keepaliveSecs from the thing

that actually knows whether work is happening.

The hub is unchanged; the operating contract is not. Serving several clusters needs no different image, chart or mode — which is exactly why the three requirements above are easy to miss until something breaks. **An address only the hub cluster can route is not an address.** Remote consumers mount `spec.service.advertiseAddress` verbatim; everything the operator can derive on its own is in-cluster-only. Deriving harder is not the answer — being told is. **Address uniqueness is not arbitrated.** Two shares may advertise the same address and both report `Conflict=False (Unique)`: that condition covers bucket-prefix overlap, so it is a true statement about storage and says nothing at all about addresses. **The failure mode is silent by construction.** A colliding client gets no error — the protocol says it is the same client returning, so the server does the correct thing with a false premise and the loser's state is simply gone.